

● EASY

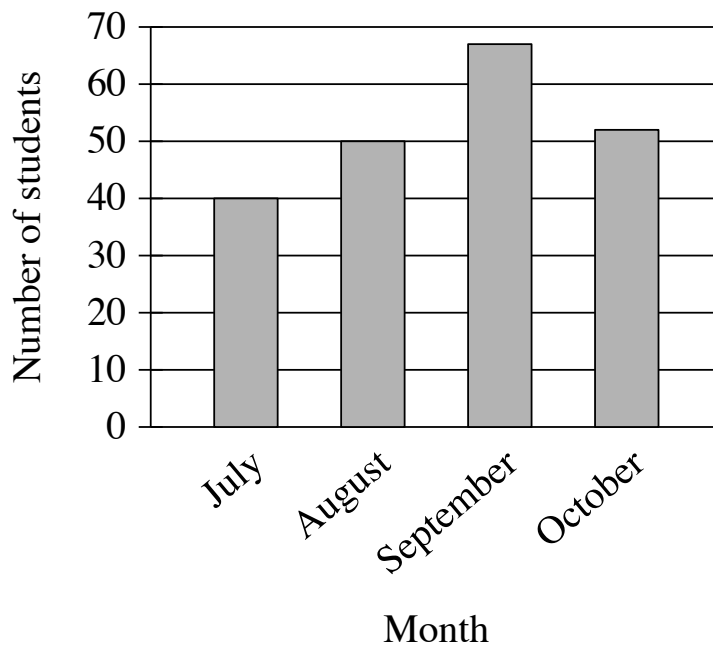
● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

15

10 questions · ~15 min

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



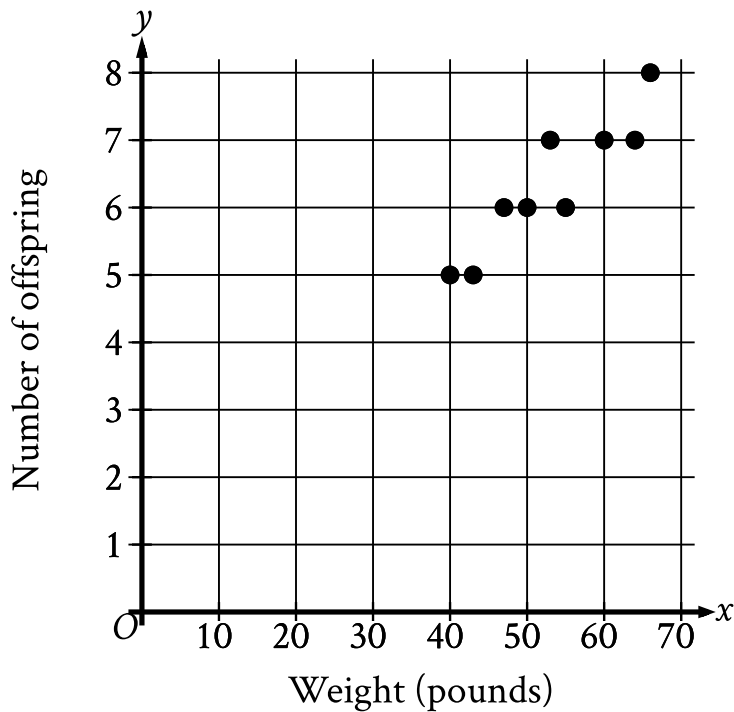
- The Number of students data for the 4 bars are as follows:
 - July: 40
 - August: 50
 - September: 67
 - October: 52

How many more students were born in August than were born in July?

- A. 90
- B. 50
- C. 40

D. 10

The scatterplot shows the relationship between the weight, in pounds, of each of 9 female gray wolves on April 30 and the number of offspring each gray wolf produced.



- The scatterplot has 9 data points.
- The data points are in a linear pattern trending up from left to right.
- The data points have the following coordinates:
 - (40 comma 5)
 - (43 comma 5)
 - (47 comma 6)
 - (50 comma 6)
 - (53 comma 7)
 - (55 comma 6)
 - (60 comma 7)
 - (64 comma 7)
 - (66 comma 8)

How many offspring did the 50-pound gray wolf produce?

A. 8

B. 7

C. 6

D. 5

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 77 rods? $5.5 \text{ yards} = 1 \text{ rod}$

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

Lani spent 15% of her 8-hour workday in meetings. How many minutes of her workday did she spend in meetings?

- A. 1.2
- B. 15
- C. 48
- D. 72

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A list of 10 data values is shown.

6, 8, 16, 4, 17, 26, 8, 5, 5, 5

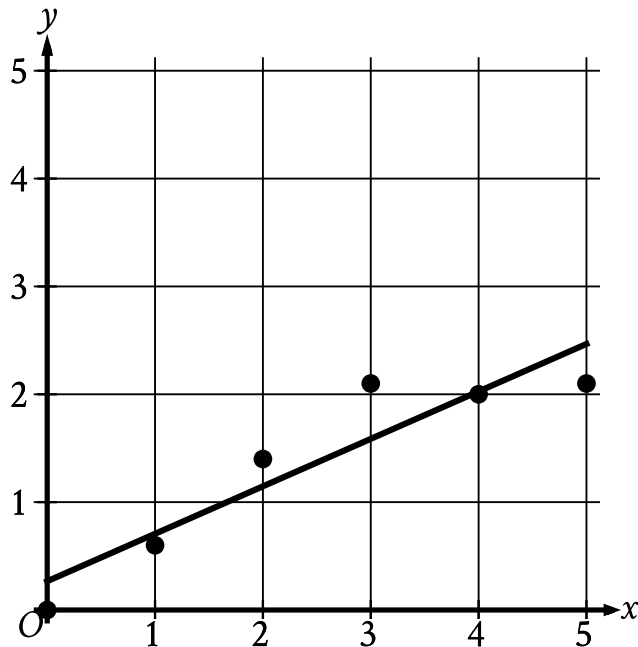
What is the mean of these data?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



- The scatterplot has 6 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - 1 point is touching the line of best fit.
 - 2 points are above the line of best fit.
 - 3 points are below the line of best fit.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 0.2)
 - (4 comma 2)

Which of the following is closest to the slope of the line of best fit shown?

A. -2.27

B. -0.44

C. 0.44

D. 2.27

To make a bakery's signature chocolate muffins, a baker needs 2.5 ounces of chocolate for each muffin. How many pounds of chocolate are needed to make 48 signature chocolate muffins? (1 pound = 16 ounces)

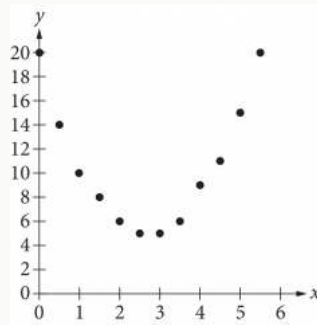
- A. 7.5
- B. 10
- C. 50.5
- D. 120

6, 10, 13, 2, 15, 22, 10, 4, 4, 4

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

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Of the following, which is the best model for the data in the scatterplot?

- A. $y = 2x^2 - 11x - 20$
- B. $y = 2x^2 - 11x + 20$
- C. $y = 2x^2 - 5x - 3$
- D. $y = 2x^2 - 5x + 3$

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A mechanical device in a workshop produces items at a constant rate of 60 items per hour. At this rate, how many items will the mechanical device produce in 3 hours?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.